ARYAN MEENA

Data Scientist | Python | SQL | Machine Learning | Quantitative Research
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EXPERIENCE

Data Scientist (Research Assistantship) | Boston University Sep 2025 – May 2026

- Traced a 15-year trading signal to biased opening prices in a free data feed by checking it against exchange data (Sharpe -1.36 vs +0.25), and the research paper is now **published in Risks (MDPI)**.
- Forecast 5-day currency volatility out of sample at R^2 0.25 vs a 0.05 noise ceiling across 8 CME futures time series, testing 6 statistical models and machine learning methods under purged walk-forward validation.
- Screened 37 financial price feeds with automated data quality checks and flagged **10 faulty series** early.
- Rewrote the Python backtest engine in NumPy to run 1,312 configurations 17x faster with identical results.
- Wrote the SQL and Python behind a live 12-table dashboard where every chart is a query anyone can rerun.

Graduate Teaching Assistant | Boston University Jan 2026 – May 2026

- Coached 60+ graduate students training as data scientists in Advanced Machine Learning and Data Science with Python, explaining model results in plain language and working through their code bugs.

FX Trader, Funded Account (Remote) | Funding Pips Dec 2023 – Dec 2024

- Hit a profit target of at least 15% of account size 3 times in a row to earn release of real trading capital on a funded prop-firm account, never breaching the 10% maximum or 5% daily drawdown limit.

Machine Learning Research Intern | ISRO Space Applications Centre Jun 2022 – Jul 2022

- Trained U-Net and DeepLabV3+ deep learning models in PyTorch to segment roads in 6,723 aerial images.

PROJECTS

Hospital Readmission Risk | Python, LightGBM, Tableau Care Dashboard

- Beat a published benchmark (AUC 0.689 vs 0.664) with predictive models on 101,766 diabetes encounters, picking LightGBM over XGBoost and logistic regression under patient-grouped cross-validation.
- Ranked patients by risk so the top 5% held **3.0x** the readmissions of a random 5%, shown in a Tableau dashboard.

Service Operations Analytics | Python, SQL, AWS, GitHub Actions Live Dashboard

- Automated a daily AWS reporting pipeline over 130K+ records that publishes only after **17 data checks** pass.
- Designed the pipeline on S3, Step Functions, Glue (PySpark) and Athena so the last good report stays live if a run breaks, and shipped it under Git version control with 81 automated tests in CI/CD.
- Picked the forecasting model by A/B testing: a challenger must cut error 2%+ in 2 of 3 windows to win.

WikiFlow: Distributed Machine Learning | PySpark, GCP Dataproc, BigQuery Paper

- Drafted the research proposal, then ran a big data job (60M Wikipedia edits, ~25 GB) on a 5-node Spark cluster on Google Cloud for about $2.50 to predict which of 1.16M new editors drop out.
- Reached AUC 0.909 with a distributed neural network, 6.4 points above logistic regression (Springer).

EDUCATION

Master of Science (MS), Applied Data Analytics, Boston University (GPA 3.75/4.0) May 2026

- Led a six-person team to a BU MedAI Hackathon podium among 40+ teams, and first-authored 5 publications.
- Coursework: Data Mining, Foundations of Machine Learning, Big Data Analytics, Data Analytics and Visualization.

Bachelor of Technology, Electronics and Communication Engineering, IIST, India May 2023

- Coursework: Statistics and Probability, Computer Vision, Machine Learning for Signal Processing.

SKILLS

Programming: Python (pandas, NumPy, scikit-learn), SQL, R, PyTorch, Spark (PySpark), Git, Docker, Excel
Machine learning: statistics, regression, classification, time series forecasting, hypothesis testing, A/B testing
Data: data cleaning, data validation, Tableau, Power BI, dashboards, AWS (S3, Glue, Athena), GCP (BigQuery)